Zulqarnain Haider

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Research Interests

Core Areas: Distributed Systems for Machine Learning, Efficient Deep Learning (MoE, PEFT, Quantization), Graph Neural Networks (GNNs), LLM Reasoning & Agentic Systems, Low-Level Concurrency & Performance Profiling.

Education

University of Engineering and Technology (UET) — Lahore, Pakistan
Master of Science in Computer Science — *Aug 2025 – Present*

FAST National University of Computer and Emerging Sciences — Islamabad, Pakistan
Bachelor of Science in Computer Science — *Aug 2019 – Jun 2023*

Research & Systems Experience

Parallel Computing Networks Lab — Islamabad, Pakistan
AI Systems Research Assistant — *Nov 2023 – Nov 2024*

- Investigated relational inductive biases by training Graph Neural Networks (GNNs) over large-scale news graph corpora in PyTorch, analyzing node and topological representations.
- Conducted research on parameter-efficient fine-tuning (PEFT/LoRA) for transformer models on bias detection benchmarks, boosting classification F1-score by 9% while tuning <1% of model weights.
- Profiled distributed training bottlenecks across multi-GPU nodes using PyTorch Distributed Data Parallel (DDP) and Ray, accelerating convergence throughput by 25% by mitigating memory and bandwidth contention.
- Designed distributed feature engineering pipelines in Apache Spark and SQL, reducing cluster I/O wait latency by 28% on high-throughput data streams.
- Constructed reproducible experimental workflows with Kubeflow and Airflow to track provenance and guarantee reproducibility across large-scale model evaluation sweeps.

CodeNinja Inc. — Lahore, Pakistan
Machine Learning Systems Engineer (Generative AI & Agents) — *Dec 2024 – Apr 2026*

- Formulated state graph transition semantics and autonomous agent planning frameworks using LangGraph and Model Context Protocol (MCP), studying systematic error recovery during tool execution.
- Investigated hybrid retrieval paradigms combining dense semantic embeddings with cross-encoder re-ranking on Qdrant and pgvector, reducing factual hallucination rates by 35%.
- Evaluated Vision Transformers (ViT) and CNN baselines (ResNet-34) on annotated datasets, conducting non-linear separability and error analysis using polynomial and RBF kernel SVMs.
- Benchmarked open-source foundation models (Llama 3, Mistral, Qwen) using PEFT/LoRA and vLLM, profiling KV-cache reuse, batching dynamics, and inference latency under concurrent load.

Replit — San Francisco, USA (Remote)
Systems Software Engineer (Low-Level Systems / Rust) — *Aug 2022 – Dec 2023*

- Developed asynchronous, high-throughput systems infrastructure in Rust (Tokio), analyzing async lock contention, thread synchronization overhead, and memory efficiency under heavy telemetry load.
- Conducted hardware-level CPU and memory profiling on multi-threaded runtimes, identifying critical hot paths and reducing end-to-end execution latency by 30%.
- Engineered connection-pooling proxies and indexing schemes in PostgreSQL, benchmarking performance boundaries to guarantee ACID compliance during concurrent transaction spikes.

Research Projects

Interpreting and Steering Spatial Reasoning via Multimodal SAEs — Academic Research
Mechanistic Interpretability & VLMs — *2026*

- Proposed a mechanistic interpretability framework utilizing Top-K Sparse Autoencoders (SAEs) to decouple spatial relational geometry from visual semantics in Vision-Language Models (LLaVA, PaliGemma).
- Engineered a PyTorch activation extraction pipeline to decompose superposed residual streams into monosemantic primitives, enabling targeted causal steering via latent clamping.
- Achieved 89.1% Target Spatial Accuracy on counterfactual benchmarks while preserving 94.7% of non-target attributes, significantly mitigating collateral semantic drift compared to linear probing.

RL-Driven Adaptive Sensing and Cascaded Inference for Edge IoT — Academic Research
TinyML & Reinforcement Learning — *2025 – 2026*

- Formulated a joint optimization framework using Deep Q-Networks (DQN) to dynamically modulate temporal sampling rates and neural network complexity on extreme-edge microcontrollers.

- Designed a cascaded inference architecture transitioning from lightweight shallow autoencoders (100 Hz) to deep 1D-CNNs (4,000 Hz) for industrial vibration anomaly detection.
- Optimized a dual energy model incorporating MEMS sensor hardware and MCU compute, strictly enforcing zero False Negative Rates (FNR) under severe industrial class imbalance.

Empirical Analysis of Non-Linear Separability in Vision Representations Academic Project
2023

- Conducted empirical comparative study analyzing representation boundaries of ViTs versus CNNs using kernelized SVM projections across varying sample complexities.

Teaching & Mentoring Experience

ReDI School of Digital Integration Volunteer
Computer Science & Python Instructor *Feb 2022 – Nov 2024*

- Delivered engaging foundational curriculum on computer science principles and introductory Python programming to diverse student cohorts.
- Designed and administered evaluation metrics and assessments to accurately track student learning progress and dynamically adapt instructional methods.
- Fostered an inclusive and supportive learning environment through personalized academic guidance and targeted conceptual support, addressing individual learning needs.
- Consistently recognized by peers and students for creating a highly supportive and engaging academic environment.

Technical & Methodological Proficiencies

ML Methods & Architectures: Mixture of Experts (MoE), Graph Neural Networks (GNNs), Vision Transformers (ViT), PEFT/LoRA, Retrieval-Augmented Generation (RAG), Agentic Reasoning & Planning
Frameworks & Libraries: PyTorch, PyTorch DDP, Hugging Face (Transformers, PEFT), Ray, Scikit-learn, vLLM, Triton Inference Server, LangGraph
Languages & Systems: Python, Rust (Tokio, async concurrency), C/C++, SQL, Bash, Linux Performance Profiling
Distributed & MLOps Tools: Kubeflow, Docker, Kubernetes, Apache Spark, Apache Airflow, PostgreSQL (pgvector), Qdrant, AWS (SageMaker, EKS)